Handout: Dual Geometry of Wavelength Space and Frequency Space — All 16 Papers + Survey (v2, fully published 2026-06-12)

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In one page

Three assumptions — $\nu\lambda = 1$ **(one relation), the zero point ½ (one constant), one asymmetric bit** — plus two operating principles (configuration reading; the record principle). From this inventory alone, the series tests how far quantum-looking and spacetime-looking structures follow **as theorems**, in a discrete system that is machine-checkable at every step. Sixteen papers plus a survey, bilingual Japanese/English, published on Zenodo. **No physical identification is made** (the central thesis: a different map with the same results).

Series map

Part	Papers	Content
Foundations	1–4	Lattice and counting: N_0(R)=1, 9, 137, … (pure enumeration); splitting; the hierarchical vacuum
Theorems	5–12	Dictionary theorem; freezing theorem (time's condition = the one bit); exclusion as theorem; area law; half-wavelength censorship; emergent gauge; **necessity of dimension 4**; measurement = appending a record
Sequel	13–16	**Resolution of position and time**; time direction = complex structure; constructed & verified coordinate map; **reduction of the measure problem**
Survey	—	The dictionary to standard theory and the **axiom exchange balance**

Main results of the sequel (published 2026-06-12)

Paper 13 — Position Had Never Been Read: discrete position = the wave's sign sector (zero additions to the state space). Space = the Pontryagin dual of the conserved-quantity lattice (its three symmetries are theorems of duality). Three readout strata $11 \subset 13 \subset 21$ (intensity-only fringes lose about half the information in principle). Clock targets = **all odd integers — proven via Gauss's triangular-number theorem**. t is a derived quantity (no independent inverse map).

Paper 14 — Time Direction and the Stage of Amplitudes: choosing a time direction = choosing a complex structure (exactly 16 lattice-compatible structures, a single B_4 orbit). The transport phase is derived and lands in Z_4 **without exception**. Symmetrization postulate → **canonical-convention theorem** (a consequence of set structure).

Paper 15 — Projection onto xyztRQ: explicit forward and inverse maps, state → (x,y,z,t,R,Q), built as a five-stage operation; **round trip 500/500; commutes with the conservation laws** (charge-like parity = a theorem of support arithmetic: 39.2% of raw combinations are violation candidates, yet zero violations among contributing paths). Velocity exists only in the sequence of records.

Paper 16 — The Rigorous Reduction of the Measure Problem: **no aggregation rule is selected.** The historical proximity 0.0007 = an artifact of two-outcome compression (with three outcomes the divergence jumps three orders). Interference granularity is not a convention: the mirror parent yields $W = 0$ **(exact)** $\wedge$ $W' = 768$ **(exact)**. The remaining freedom reduces to **granularity × one event bit × conditioning**, with a table of adjudication experiments installed inside the model. The Born rule = "an effective law to merge with in the appropriate limit" (triple criterion: convergence, suppression, prediction).

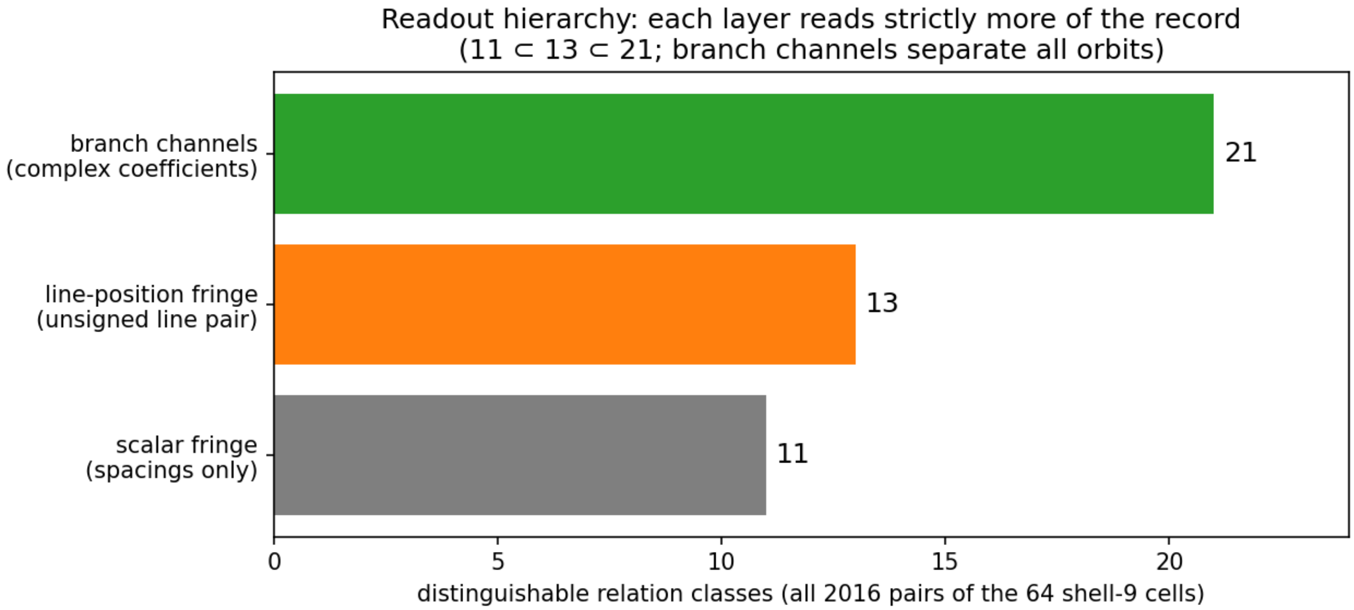

Figure 1: The readout hierarchy 11 in 13 in 21

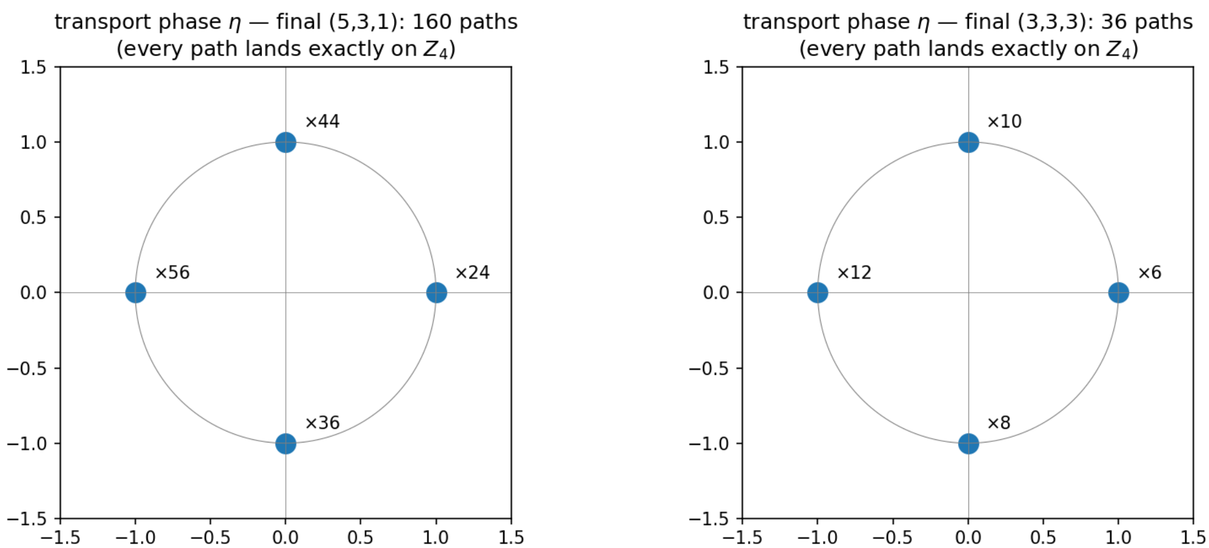

Figure 2: Transport phases land exactly on Z4

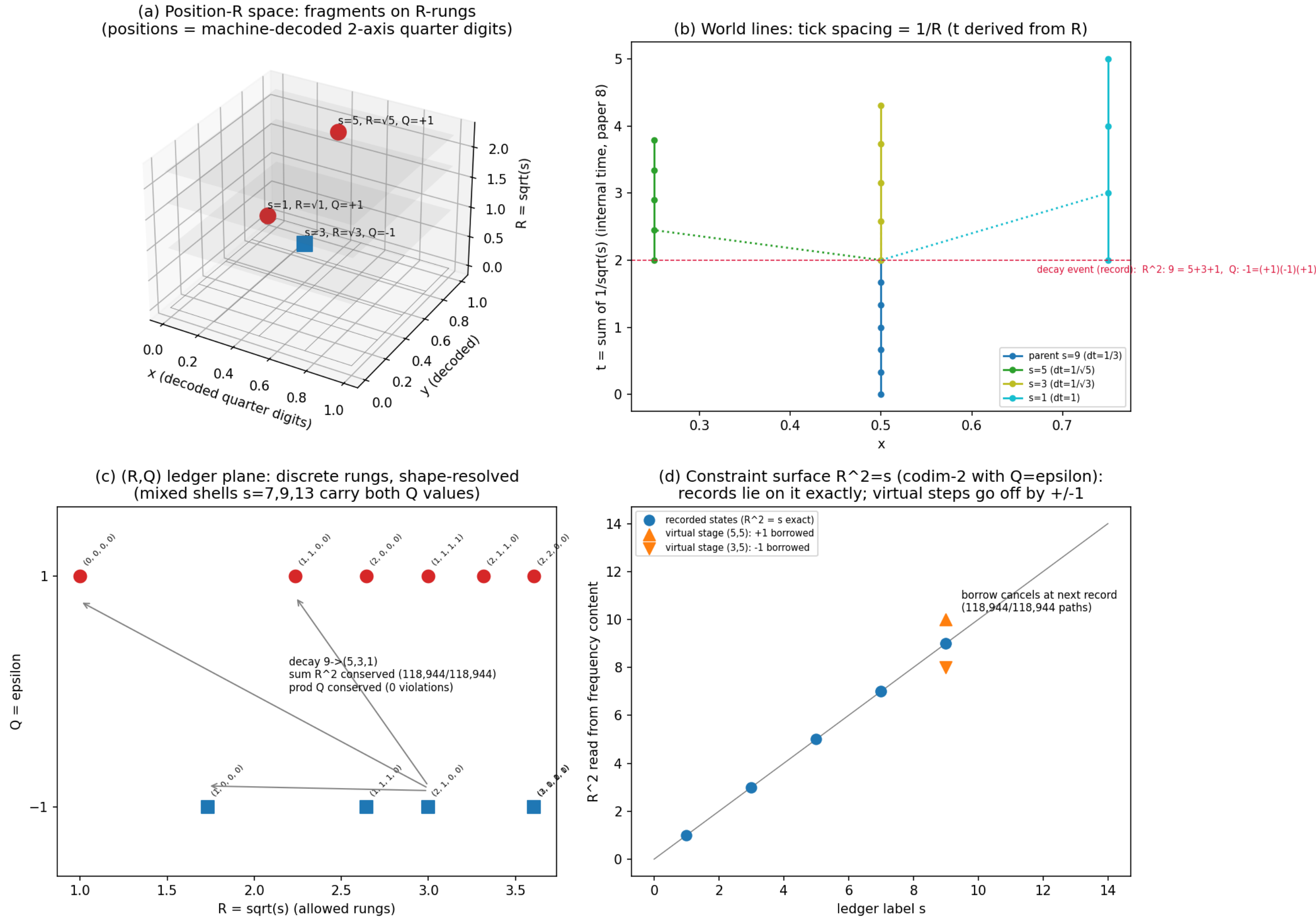

Figure 3: The xyztRQ space at a glance

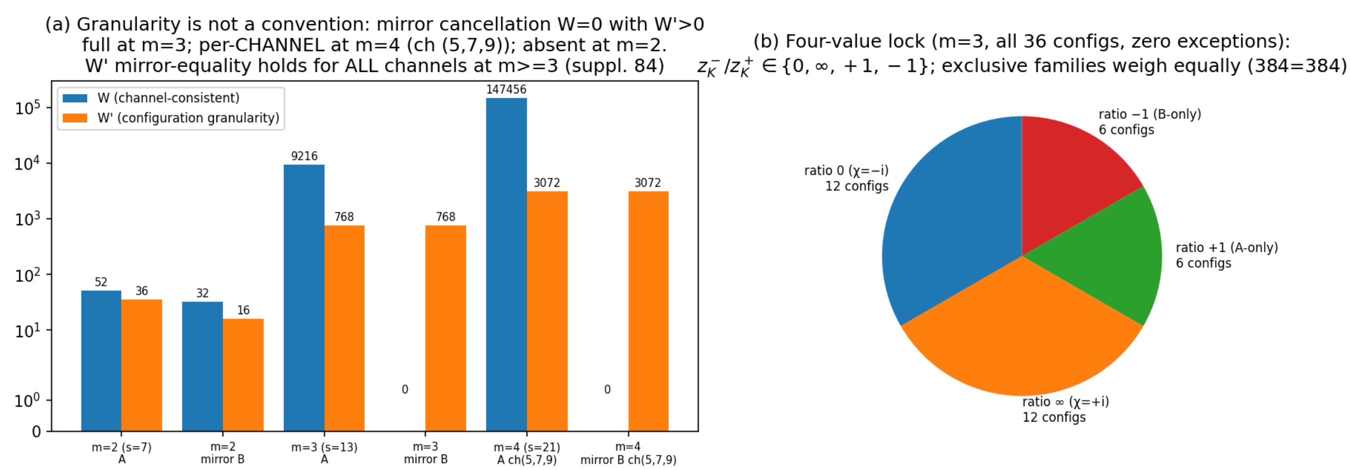

Figure 4: Granularity is not a convention; the four-value lock

Top to bottom: the readout hierarchy (intensity 11 $\subset$ line-position 13 $\subset$ branch channels 21 = complete separation; Paper 13) / transport phases landing exactly on Z_4 (all 196 paths, zero exceptions; Paper 14) / **the xyztRQ space at a glance** (3D arrangement, world lines with tick $1/R$, the (R, Q) ledger plane, the constraint surface $R^2 = s$ with virtual ±1; Paper 15) / the mirror cancellation $W = 0 \wedge W' > 0$ and the four-value lock (Paper 16). All figures are exact computations.

The survey: the axiom exchange balance

Axiom in standard theory	In this system	Source
Symmetrization postulate	**Theorem** (canonical convention)	Paper 14
Time = external parameter	**Theorem** (derived quantity)	Papers 13, 15
Measurement postulate	**Theorem** (appending + record)	Paper 12
Spacetime dimension = input	**Theorem** (necessity)	Paper 11
Charge conservation	**Theorem** (support parity arithmetic)	Paper 15

Payment: one relation + one constant + one bit + two operating principles. **Only two entries remain blank: measure and action.** Thesis: a completely filled dictionary = re-notation = zero empirical content. **The incomplete points of the connection are the only places where observation can adjudicate** — the remainder is at once a deficiency and a map of decidability.

Verification regime

Every claim is supported by exhaustive verification, exact identities, and machine-precision numerics; all figures are exact computations (no schematics). Papers 13–16 passed a **two-party AI independent-verification protocol** (local machine verification × independent re-computation and review): **two rounds each, eight reviews in total**, with seven mutual corrections (including refuted predictions and retractions) — **the history is published undeleted**.

Access

- **Papers 1–16**: Zenodo Concept DOIs 10.5281/zenodo.20588036 – 20665699 (full list in the paper index, Ronbun-Ichiran.md, in the series folder)
- **Survey**: 10.5281/zenodo.20666132
- **Repository snapshot** (permanent archive of everything): 10.5281/zenodo.20666114
- **Repository** (84 supplements, all scripts, review records, indexes): https://github.com/WurabeSeiji/ai-chat-logs-open (series folder: "Dual Geometry of Wavelength and Frequency Spaces" — Japanese folder name; identifiable as the only series folder)

Standing (restated)

Not an attempt at a new or unified theory. **"Structures close to standard quantum theory may be derivable from a different set of simple assumptions"** — that is the prescribed reading. $N_0(3)=137$ is not identified with the fine-structure constant. No measure has been selected. Numerical conversion to physical units would require an independent dimensionless observable (stated as unresolved).